

EMBEDDED SYSTEMS

6-Month Professional Training Program • 70% Practical + 30% Theory

Program	Embedded Systems Professional Training
Duration	6 Months
Approach	70% Practical + 30% Theory
Languages	Embedded C / C++
Microcontrollers	8051, Arduino Uno, PIC16F877A
Tools	Keil, Proteus, Arduino IDE, Wokwi, MPLAB X, XC8, Git/GitHub
Training Style	Hands-on Labs + Mini Projects + Weekly Assessment + Final Project
Eligibility	Diploma, BE/B.Tech, ME/M.Tech, BSc/MSc in Electronics

Course Learning Path

- 01 Month 1 — Embedded C & 8051 Fundamentals
- 02 Month 2 — 8051 Interfacing & Peripherals
- 03 Month 3 — Arduino Fundamentals & Hardware Interfacing
- 04 Month 4 — Communication Protocols & Arduino Projects
- 05 Month 5 — PIC Microcontroller
- 06 Month 6 — Advanced Concepts & Final Project

Program Structure

- Foundation: Embedded Systems concepts, Embedded C, registers, memory and microcontroller architecture.
- Interfacing: Digital/analog I/O, displays, keypads, timers, interrupts, UART and sensors.
- Rapid Prototyping: Arduino programming, Wokwi simulation, sensors and actuators.
- Communication: UART, I2C, SPI and CAN fundamentals with practical applications.
- Microcontroller Development: PIC16F877A firmware using MPLAB X, XC8 and Proteus.
- Industry-Oriented Integration: debugging, datasheet reading, RTOS concepts, Git/GitHub and final project development.

Training Outcome

By the end of the program, students should be able to design, program, simulate, test, debug and document practical embedded-system applications using multiple microcontroller platforms.

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Month 1 — Embedded C & 8051 Fundamentals

Module	01 — Embedded C & 8051 Fundamentals
Goal	Build a strong foundation in Embedded C programming, microcontroller concepts, registers, memory, and 8051 architecture.
Level	Beginner to Intermediate
Main Focus	Theory fundamentals + practical implementation + mini project
Tools	Keil + Proteus + 8051 Development Board

Module 01 Learning Path

- Introduction to Embedded Systems
- Microcontroller vs Microprocessor
- Embedded C Fundamentals
- Variables, Data Types and Operators
- Conditional Statements and Loops
- Functions, Arrays and Strings
- Pointers and Structures
- Bitwise Operations
- Registers and Memory Concepts
- 8051 Architecture and I/O Ports

Practical Labs

- LED Blinking and LED Patterns
- Push Button and Buzzer Interfacing
- 7-Segment Display Interfacing
- Switch-Based Applications

Mini Projects

- LED Pattern Controller
- Switch-Controlled Buzzer System

Module Completion Checklist

Requirement	Complete
Understand the module theory concepts	■
Complete the listed practical laboratory exercises	■
Verify the circuit/application in the selected tool or hardware	■
Complete at least one mini project	■
Document the implementation and observations	■
Complete the weekly assignment/test requirements	■

Module Outcome

Students can apply the module concepts in practical embedded-system development, simulation, testing and debugging.

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Month 2 — 8051 Interfacing & Peripherals

Module	02 — 8051 Interfacing & Peripherals
Goal	Develop practical skills in 8051 peripherals, timing, interrupts, serial communication, ADC/PWM fundamentals, and sensor interfacing.
Level	Beginner to Intermediate
Main Focus	Theory fundamentals + practical implementation + mini project
Tools	Keil + Proteus + 8051 Hardware

Module 02 Learning Path

- LCD and Keypad Interfacing
- Timers and Counters
- Interrupts
- UART / Serial Communication
- ADC and PWM Fundamentals
- Sensor Interfacing

Practical Labs

- LCD Display Application
- 4×4 Keypad Interface
- Timer/Counter Application
- Interrupt-Based Input Handling
- UART Serial Communication
- Sensor Data Display

Mini Projects

- Digital Counter and Digital Clock
- Password-Based Security System
- Temperature Monitoring System
- Serial Communication Application

Module Completion Checklist

Requirement	Complete
Understand the module theory concepts	■
Complete the listed practical laboratory exercises	■
Verify the circuit/application in the selected tool or hardware	■
Complete at least one mini project	■
Document the implementation and observations	■
Complete the weekly assignment/test requirements	■

Module Outcome

Students can apply the module concepts in practical embedded-system development, simulation, testing and debugging.

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Month 3 — Arduino Fundamentals & Hardware Interfacing

Module	03 — Arduino Fundamentals & Hardware Interfacing
Goal	Learn Arduino Uno architecture, Arduino IDE programming, I/O, sensors, actuators, PWM, and rapid prototyping.
Level	Beginner to Intermediate
Main Focus	Theory fundamentals + practical implementation + mini project
Tools	Arduino Uno + Arduino IDE + Wokwi

Module 03 Learning Path

- Arduino Architecture and Arduino Uno
- Arduino IDE, Digital and Analog I/O
- Serial Communication and PWM
- Functions and Libraries
- Sensors and Actuators

Practical Labs

- LEDs, RGB LEDs, Push Buttons and Potentiometers
- LDR and Temperature Sensor
- IR Sensor
- Buzzer, Servo Motor, DC Motor and Relay
- Ultrasonic Sensor

Mini Projects

- Automatic Street Light
- Obstacle Detection System
- Temperature Monitoring System

Module Completion Checklist

Requirement	Complete
Understand the module theory concepts	■
Complete the listed practical laboratory exercises	■
Verify the circuit/application in the selected tool or hardware	■
Complete at least one mini project	■
Document the implementation and observations	■
Complete the weekly assignment/test requirements	■

Module Outcome

Students can apply the module concepts in practical embedded-system development, simulation, testing and debugging.

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Month 4 — Communication Protocols & Arduino Projects

Module	04 — Communication Protocols & Arduino Projects
Goal	Understand common embedded communication protocols and use Arduino to build multi-device communication projects.
Level	Beginner to Intermediate
Main Focus	Theory fundamentals + practical implementation + mini project
Tools	Arduino IDE + Hardware Modules

Module 04 Learning Path

- UART: TX/RX, Baud Rate, Data Transmission and Reception
- I2C: Master-Slave Concept, SDA/SCL, Addressing
- SPI: MOSI, MISO, SCLK, CS/SS
- CAN Basics: Bus Architecture, Nodes and Messages
- Automotive Applications of CAN
- Multi-Device Communication

Practical Labs

- UART Communication
- I2C LCD/OLED Interfacing
- I2C Sensor Interfacing
- SPI Device Interfacing
- Multi-Device Communication

Mini Projects

- Smart Home Automation
- Sensor Monitoring System
- Serial-Controlled Device

Module Completion Checklist

Requirement	Complete
Understand the module theory concepts	■
Complete the listed practical laboratory exercises	■
Verify the circuit/application in the selected tool or hardware	■
Complete at least one mini project	■
Document the implementation and observations	■
Complete the weekly assignment/test requirements	■

Module Outcome

Students can apply the module concepts in practical embedded-system development, simulation, testing and debugging.

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Month 5 — PIC Microcontroller

Module	05 — PIC Microcontroller
Goal	Develop practical PIC16F877A programming and interfacing skills using MPLAB X, XC8, Proteus, and a PIC development board.
Level	Beginner to Intermediate
Main Focus	Theory fundamentals + practical implementation + mini project
Tools	MPLAB X IDE + XC8 + Proteus + PIC Development Board

Module 05 Learning Path

- PIC Architecture — PIC16F877A
- GPIO and Registers
- ADC and Timers
- Interrupts
- PWM
- LCD and Keypad
- UART
- Sensor Interfacing

Practical Labs

- LED, Switch and LCD Interfacing
- ADC-Based Sensor Reading and Temperature Monitoring
- Motor Control
- UART Communication
- PWM Applications

Mini Projects

- ADC-Based Temperature Monitor
- PWM Motor Controller
- UART Sensor Monitor

Module Completion Checklist

Requirement	Complete
Understand the module theory concepts	■
Complete the listed practical laboratory exercises	■
Verify the circuit/application in the selected tool or hardware	■
Complete at least one mini project	■
Document the implementation and observations	■
Complete the weekly assignment/test requirements	■

Module Outcome

Students can apply the module concepts in practical embedded-system development, simulation, testing and debugging.

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Month 6 — Advanced Concepts & Final Project

Module	06 — Advanced Concepts & Final Project
Goal	Integrate microcontrollers, peripherals, communication protocols, debugging, documentation, and project development into a complete embedded system.
Level	Beginner to Intermediate
Main Focus	Theory fundamentals + practical implementation + mini project
Tools	All Previous Tools + Git/GitHub

Module 06 Learning Path

- Inter-Microcontroller Communication — UART / I2C / SPI
- CAN Fundamentals and Interrupt-Driven Programming
- Timers, PWM, ADC and Sensor Integration
- Debugging Techniques and Datasheet Reading
- Basics of Embedded System Design
- Introduction to RTOS Concepts
- Git/GitHub Basics

Practical Labs

- Inter-Microcontroller Communication
- Interrupt-Driven Sensor Application
- Multi-Sensor Integration
- Protocol-Based Device Control
- Debugging and Datasheet Exercise
- Git/GitHub Project Workflow

Mini Projects

- Smart Home Automation System
- IoT-Based Temperature Monitoring System
- Smart Parking System
- Obstacle Detection and Alert System
- Automated Street Light System
- Industrial Temperature and Safety Monitoring System

Module Completion Checklist

Requirement	Complete
Understand the module theory concepts	■
Complete the listed practical laboratory exercises	■
Verify the circuit/application in the selected tool or hardware	■
Complete at least one mini project	■
Document the implementation and observations	■
Complete the weekly assignment/test requirements	■

Module Outcome

Students can apply the module concepts in practical embedded-system development, simulation, testing and debugging.

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Final Project & Assessment

Final Project — Complete Embedded System

Students will develop a complete embedded system integrating a microcontroller, Embedded C, sensors, actuators, display, communication protocols, and interrupts/timers.

Suggested Projects

- Smart Home Automation System
- IoT-Based Temperature Monitoring System
- Smart Parking System
- Obstacle Detection and Alert System
- Automated Street Light System
- Industrial Temperature and Safety Monitoring System

Recommended Final Project Flow

Start → Requirement Definition → Block Diagram → Component Selection → Circuit Design → Firmware Development → Simulation → Hardware Integration → Debugging → Testing → Documentation → GitHub → Final Demonstration → End

Assessments

- Weekly Test: Conducted at the end of every week based on topics covered.
- Weekly Assignment: Embedded C programs, circuit simulations, microcontroller programming, sensor interfacing, debugging exercises, protocol-based tasks and mini-projects.
- Practical Evaluation: Demonstration of selected laboratory experiments and mini projects.
- Final Project Evaluation: Working demonstration, firmware quality, debugging approach, documentation and project presentation.

Overall Program Exit Criteria

Requirement	Complete
Write and understand Embedded C programs	■
Explain microcontroller architecture and memory concepts	■
Interface LEDs, switches, displays, keypads, sensors and actuators	■
Use timers, counters, ADC, PWM and interrupts	■
Implement UART communication	■
Understand and implement I2C and SPI basics	■
Explain CAN fundamentals and bus architecture	■
Develop applications using 8051, Arduino and PIC16F877A	■
Use Keil, Proteus, Arduino IDE, Wokwi, MPLAB X and XC8	■
Apply debugging techniques and read datasheets	■
Understand basic RTOS concepts	■
Use Git/GitHub for project version control	■
Complete and demonstrate a final embedded system project	■

Professional Outcome

Learn online. Build real projects. Develop practical embedded-system skills. Become industry-ready.

Institute	Embedded Electronics
Website	https://embedded-electronics.github.io/